

Binomial Random Variables

Counting the number of successes

Download the section 8 .Rmd handout to
STAT240/lecture/08-binomial.

We've seen examples of probability distributions.
How do we take a real-life process and figure out its distribution?

Real life is messy and complex. Options:

- Repeat the process infinity times
- Make some **reasonable assumptions**

What is the probability distribution of how many green lights I hit out of three? Let's assume:

- Lights are independent
- Each one has probability p of being green

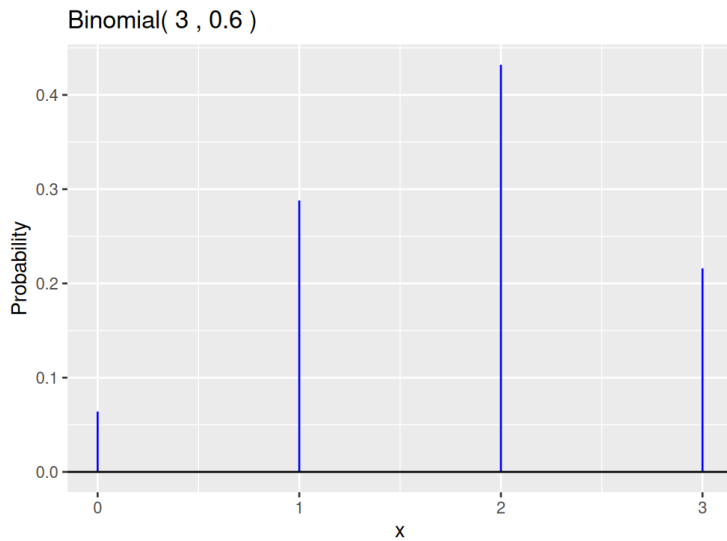
This simplification lets us calculate probabilities. Suppose we have $p = 0.6$.

X = number of green lights.

Outcomes	x	$P(X = x)$
RRR	0	$(0.4)(0.4)(0.4) = 0.064$
	1	
GGR, GRG, RGG	2	$3(0.6)(0.6)(0.4) = 0.432$
	3	

Complete the pmf for X .

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X in the traffic lights example is a count of “successes” in 3 tries.

X is a **binomial** RV. A binomial counts the number of times a desired outcome occurs in many tries.

We say X has a “binomial distribution”.

The lights (trials) have specific properties:

- **B**: they are binary (success, or failure)
- **I**: they are independent
- **N**: fixed sample size n
- **S**: they have the same probability p

A binomial RV is the count of successes in a fixed number (n) of identical trials.

What if we counted how many green lights we hit, before seeing the first red light? Not binomial.

What if we also counted yellow lights? Not binomial.

Write $X \sim \text{Binom}(n, p)$.

- n : pre-determined number of trials
- p : individual success probability

So $X \sim \text{Binom}(3, 0.6)$ for the traffic lights.

The shape of the distribution depends on n and p .

For the traffic lights,

$$P(X = 2) = 3(0.6)^2(0.4)^1$$

- 2 successes, 1 failure
- 0.6 is the success probability
- 0.4 is the failure probability
- 3 is the number of ways to have 2/3 successes.

In general,

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

- x successes, $n - x$ failures
- p is the success probability
- $1 - p$ is the failure probability
- $\binom{n}{x}$ is the number of orderings

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

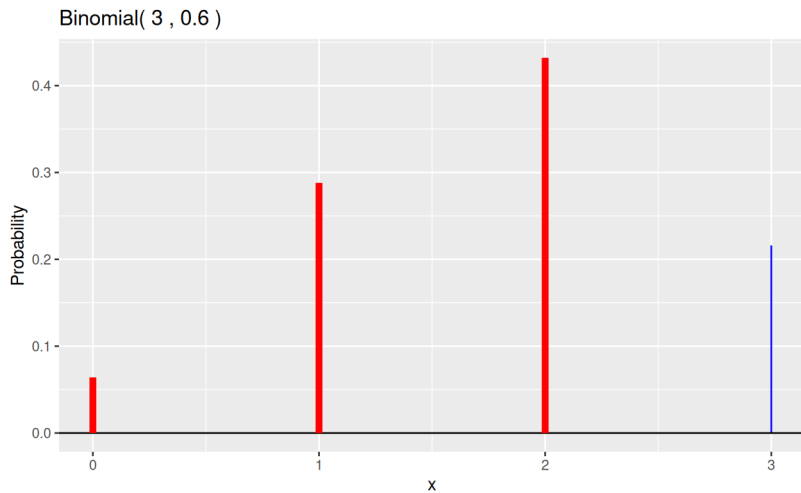
$n!$ is the product of numbers 1 to n .

Can be calculated with R's `choose` or `factorial`.

$$P(X = x) = \binom{n}{x} p^x (1 - p)^{n-x}$$

We can use the R command `dbinom` as a shortcut.

We can also use `pbinom` for a “less than or equal” (cumulative) probability.



Remember `pbinom` calculates area to the *left*, including the x value specified.

Let $Y \sim \text{Binom}(8, 0.3)$.

- What is $P(Y > 4)$?
- What is $P(Y \geq 4)$?
- What is $P(3 \leq Y \leq 6)$?

We have shortcut formulas for the mean and variance of $X \sim \text{Binom}(n, p)$.

$$\text{mean } \mu = np, \quad \text{var } \sigma^2 = np(1 - p)$$

In the lights example:

- $\mu = 3(0.6) = 1.8$
- $\sigma^2 = 3(0.6)(0.4) = 0.72$

The mean is the “average” or “typical” values of the binomial process.

Another way to quantify the RV is with a **percentile** or **quantile**.

In a population, 35% of individuals are below the 35th percentile, and 65 are above.

In general, the quantile for p subdivides the probability such that there is p probability to the left and $1 - p$ to the right.

Percentiles work a bit differently here since a binomial process is discrete. How large is q before

$$P(\text{Binom} \leq q)$$

is at least p ?

We can find binomial quantiles with `qbinom`. This is the inverse of `pbinom`.

- `pbinom`: input x value, output cumulative probability
- `qbinom`: input cumulative probability, output x value

Command	In	Out
<code>dbinom</code>	A value x	$P(X = x)$
<code>pbinom</code>	A value x	$P(X \leq x)$
<code>qbinom</code>	A probability p	q for $P(X \leq q) = p$

Let $X \sim \text{Binom}(90, 0.7)$.

- Find the mean μ and sd σ of X
- What is $P(X = \mu)$?
- What is $P(\mu - \sigma \leq X \leq \mu + \sigma)$?
- What are the 5th and 95th percentiles of X ?